

SEAN MICHAEL KELLY

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PROFESSIONAL SUMMARY

Machine Learning Engineer with 7 years of experience building and deploying production ML systems across fintech and healthcare sectors. Strong background in Python development, deep learning frameworks, and cloud infrastructure. I've led teams through complex model training pipelines and mentored junior engineers on best practices for model validation and monitoring. Looking to contribute to challenging problems where machine learning can solve real business problems.

TECHNICAL SKILLS

Programming Languages: Python, SQL, Java, Scala

Machine Learning: TensorFlow, PyTorch, Scikit-learn, XGBoost, LightGBM

Data Engineering: Apache Spark, Pandas, NumPy, Dask

Cloud Platforms: AWS (SageMaker, EC2, S3), Google Cloud Platform, Azure ML

Deep Learning: Neural Networks, CNN, RNN, Transformers, BERT

Databases: PostgreSQL, MongoDB, DynamoDB, Elasticsearch

DevOps and MLOps: Docker, Kubernetes, Git, Jenkins, MLflow, Airflow

Other: REST APIs, A/B Testing, Statistical Analysis, Tableau

WORK EXPERIENCE

Senior Machine Learning Engineer

FinScale Analytics

Dublin, Ireland

finscale.io

January 2022 - Present

Designed and implemented automated fraud detection system using gradient boosting that reduced false positives by 34 percent while maintaining 99.2 percent recall across 2.3 million daily transactions.

Led a team of three junior ML engineers on a credit risk modeling project, providing code reviews and helping debug model drift issues that were causing prediction accuracy to decline by 2 percent monthly.

Migrated legacy model serving infrastructure from on-premise servers to AWS SageMaker, cutting deployment time from 6 hours to 15 minutes and reducing infrastructure costs by 40 percent annually.

Established MLOps practices including automated model retraining pipelines, monitoring dashboards, and documentation standards that improved team velocity and reduced production incidents by 60 percent.

Machine Learning Engineer

MediPredict Healthcare

Cork, Ireland

medipredict.co

May 2019 - December 2021

Built patient readmission prediction models using RNN architectures on sequential clinical data, achieving 87 percent AUC and helping hospital reduce preventable readmissions by 12 percent in pilot.

Developed feature engineering pipelines to extract meaningful signals from unstructured patient notes using NLP techniques, improving model performance by 8 percent on downstream prediction tasks.

Collaborated with clinical stakeholders and product team to define success metrics and translate business requirements into machine learning objectives for three major model releases.

Managed model lifecycle from experimentation to production monitoring, implementing automated retraining triggers and performance dashboards that caught data drift issues before they impacted clinical decisions.

Junior Machine Learning Engineer

DataFlow Systems

Dublin, Ireland

dataflow.net

August 2017 - April 2019

Contributed to recommendation engine serving personalized content to 500000 active users, implementing collaborative filtering and content-based approaches that increased user engagement by 18 percent.

Preprocessed and analyzed large datasets using Spark and Pandas, building data quality checks and automated validation pipelines that reduced training data issues by 50 percent.

Trained and evaluated multiple machine learning models for time series forecasting, comparing ARIMA, Prophet, and LSTM approaches to predict customer demand with 92 percent accuracy.

Participated in agile development process, writing unit tests and documentation for ML code, and helped maintain CI/CD pipelines that enabled weekly model deployments.

EDUCATION

Bachelor of Science in Computer Science

University College Dublin

Dublin, Ireland

2013 - 2017

GITHUB REPOSITORIES

github.com/seankelly-ml/fraud-detection-ensemble

Gradient boosting ensemble for credit card fraud detection with 99.2 percent recall.

github.com/seankelly-ml/clinical-nlp-pipeline

NLP feature extraction from clinical notes using transformers and regex patterns.

github.com/seankelly-ml/mlops-monitoring-kit

Python library for ML model monitoring, drift detection, and automated alerts.

github.com/seankelly-ml/recommendation-system

Collaborative filtering and content-based recommendation engine with Flask API.

CERTIFICATIONS

AWS Certified Machine Learning - Specialty

June 2021

Deep Learning Specialization (Coursera)

December 2019